

Motov - MotorBike rental management system

Mai Xuan Tung¹, Le Thi Bao Uyen¹, Dang Phuong Khoi Nguyen¹, Tran Quang Buu Hoang¹, and Nguyen Ha Tien Quang¹

FPT University, Da Nang, Vietnam group-email@fpt.edu.vn

Abstract. The rapid growth of digital platforms has transformed the transportation and rental industry. However, many motorbike rental businesses still rely on manual processes, resulting in inefficient booking management, vehicle tracking, identity verification, and payment handling. This paper proposes Motov, a hybrid motorbike rental management platform that integrates company-owned and owner-owned vehicles into a unified ecosystem. The system provides web and mobile applications supporting customers, staff, vehicle owners, and administrators. Core functionalities include vehicle catalog management, online booking, rental conflict validation, identity verification, payment processing, notification services, and administrative analytics. The proposed solution aims to improve operational efficiency, increase vehicle utilization, enhance customer experience, and support secure rental transactions. The platform is implemented using ReactJS, React Native, ASP.NET Core Web API, and MongoDB.

Keywords: Motorbike Rental · Vehicle Management · Online Booking · Identity Verification · Mobile Application

1 Introduction

The demand for motorbike rental services has increased significantly due to tourism growth, urban mobility needs, and the popularity of short-term transportation solutions. In Vietnam, many rental businesses still operate through manual processes such as phone calls, social media messaging, and paper-based agreements.

Traditional rental approaches often face several challenges, including inefficient vehicle management, limited booking transparency, and difficulties in tracking rental transactions. Furthermore, customers may experience inconvenience when searching for available vehicles, while rental businesses struggle to maximize vehicle utilization.

Recent studies have investigated vehicle rental systems, fleet management, customer satisfaction, and peer-to-peer vehicle sharing models [3, 2, 5, 4, 6, 7].

This project proposes Motov, a hybrid motorbike rental platform that supports both company-owned and partner-owned vehicles. The system allows customers to search and book vehicles online, partners to register and manage their vehicles, and administrators to oversee platform operations.

The main contributions of this project are:

- Development of a hybrid rental model integrating company-owned and owner-owned motorbikes.
- Design of a centralized rental management platform supporting customers, staff, owners, and administrators.
- Integration of identity verification, booking conflict validation, and automated rental price calculation mechanisms.
- Implementation of web and mobile applications using modern full-stack technologies.
- Improvement of booking transparency, operational efficiency, and vehicle utilization.

2 Related Work

Several studies have investigated vehicle rental, fleet management, customer satisfaction, and car-sharing systems.

Khawaja and Yang [3] examined the relationship between employee engagement, customer satisfaction, and customer retention in the car rental industry. Their findings indicated that service quality significantly influences customer loyalty and business performance. However, the study focused on customer behavior rather than software platform development.

Gonzalez et al. [2] proposed a fleet utilization metric for shared mobility systems. Although the study improved operational evaluation, it did not address booking management or payment integration.

Munzel et al. [5] analyzed different car-sharing business models and demonstrated the viability of peer-to-peer vehicle sharing. However, the research did not provide a practical software implementation.

Liu et al. [4] investigated booking optimization algorithms for online car-sharing systems. Their work improved vehicle allocation efficiency but lacked user management and payment functionalities.

Neifer et al. [6] explored trust-building mechanisms in peer-to-peer car-sharing platforms. The study highlighted the importance of reputation systems but did not provide a complete rental management solution.

Andersen [1] proposed a queueing model for rental scheduling and inventory optimization. However, the study focused on mathematical optimization rather than platform implementation.

Soppert et al. [7] studied the integration of traditional rental and car-sharing services. Their findings showed improved utilization and profitability, but no web or mobile implementation was proposed.

Although previous studies have focused on fleet utilization, scheduling optimization, customer satisfaction, and peer-to-peer sharing, few studies provide a comprehensive platform that integrates booking, payment processing, vehicle management, partner vehicle registration, and administrative monitoring.

Therefore, this project proposes a hybrid motorbike rental platform that combines company-owned and partner-owned vehicles through a centralized web and mobile ecosystem.

3 Method

This study adopts the Design Science Research (DSR) approach to develop and evaluate the Motov platform. The objective is to design an artifact that addresses practical challenges in motorbike rental operations, including fragmented booking processes, inefficient vehicle utilization, and insufficient customer verification mechanisms.

3.1 Research Framework

The proposed research process consists of five stages:

1. Problem Identification: Investigating limitations of traditional motorbike rental businesses through requirement analysis and stakeholder discussions.
2. Objective Definition: Defining system goals related to booking transparency, operational efficiency, customer convenience, and security enhancement.
3. Artifact Design: Designing the hybrid rental platform architecture, database structure, and functional modules.
4. Development and Implementation: Building the web application, mobile application, backend services, and database components.
5. Evaluation: Verifying that the implemented functionalities satisfy predefined requirements and business rules.

3.2 Development Process

The Agile Scrum methodology was adopted to support iterative development and continuous stakeholder feedback. The development activities were divided into multiple sprints, each consisting of planning, implementation, testing, review, and retrospective phases.

This approach enabled the development team to continuously refine system functionalities and rapidly respond to requirement changes throughout the project lifecycle.

3.3 Proposed System Architecture

The proposed system adopts a three-tier architecture consisting of presentation, application, and data layers, as illustrated in Figure ??.

- Presentation Layer: ReactJS web application for administrators, staff, and owners, together with React Native mobile applications for customers.
- Application Layer: ASP.NET Core Web API responsible for business logic processing, authentication, booking validation, payment handling, and notification management.
- Data Layer: MongoDB database storing user profiles, vehicle information, bookings, payments, verification records, and system logs.

The layered architecture improves maintainability, scalability, and separation of concerns while supporting future system expansion.

3.4 Hybrid Rental Model

Unlike conventional rental systems that only support company-owned fleets, Motov introduces a hybrid rental model integrating both company-owned and owner-owned vehicles.

The proposed model allows individual vehicle owners to register their motor-bikes on the platform while administrators maintain centralized supervision over operational activities. This approach increases fleet availability without requiring proportional investments in vehicle acquisition.

The hybrid model provides benefits for multiple stakeholders:

- Customers gain access to a wider range of vehicles.
- Vehicle owners receive an additional income source.
- Administrators maintain unified management and quality control.
- The platform improves overall vehicle utilization rates.

3.5 Booking and Validation Mechanism

The booking workflow constitutes the core process of the proposed system.

When a customer submits a booking request, the system executes the following sequence:

1. Verify customer authentication status.
2. Validate customer identity verification status.
3. Check vehicle availability during the requested rental period.
4. Detect scheduling conflicts with existing reservations.
5. Calculate the rental price according to duration and promotional policies.
6. Process online or cash payment options.
7. Generate booking records and update vehicle status.
8. Dispatch confirmation notifications to relevant stakeholders.

This mechanism minimizes double-booking incidents and ensures consistency across rental transactions.

3.6 Rental Conflict Validation

To prevent overlapping reservations, the system employs a temporal validation mechanism.

For a requested booking interval (S_r, E_r) and an existing booking interval (S_e, E_e) , a conflict occurs if:

$$S_r < E_e \wedge E_r > S_e$$

If the condition evaluates to true, the booking request is rejected and alternative scheduling options must be selected.

This validation strategy guarantees that each vehicle can only participate in one active rental transaction during a specific time interval.

3.7 Rental Price Calculation

The system automatically computes rental costs using the following formula:

$$TotalPrice = (RentalDays \times DailyRate) - Discount + AdditionalFees$$

where:

- *RentalDays* represents the number of rental days;
- *DailyRate* denotes the vehicle rental fee per day;
- *Discount* refers to promotional reductions;
- *AdditionalFees* include late-return penalties or supplementary service charges.

Automated price calculation reduces human errors and improves pricing transparency.

3.8 Identity Verification Process

To improve transaction security, Motov incorporates a citizen identity verification process.

Customers are required to upload images of their Citizen Identification Card (CCCD). Administrators review submitted documents before granting verified status. Only verified customers are eligible to complete vehicle pickup procedures.

This mechanism enhances trust among stakeholders and reduces risks associated with fraudulent transactions.

3.9 Evaluation Strategy

The proposed platform is evaluated using functional testing methods based on predefined use cases and business requirements.

The evaluation focuses on the following criteria:

- Correctness of booking operations;
- Accuracy of rental conflict detection;
- Reliability of price calculation mechanisms;
- Effectiveness of identity verification workflows;
- Usability of web and mobile interfaces.

Successful completion of these evaluation criteria indicates that the proposed solution effectively addresses the operational limitations of conventional motorbike rental systems.

4 System Analysis

4.1 Problem Analysis

Many motorbike rental businesses still depend on manual workflows, including phone-based reservations, social media communication, and paper-based contracts. These approaches often lead to inefficient booking management, poor vehicle utilization, booking schedule conflicts, lack of customer identity verification, and limited operational visibility. In addition, rental businesses face difficulties in monitoring fleet availability, tracking rental transactions, and generating business reports.

4.2 Proposed Solution

The proposed solution is a centralized hybrid motorbike rental platform that supports both company-owned and owner-owned vehicles. The system provides web and mobile applications for customers, owners, staff, and administrators. It integrates online booking, payment processing, citizen identity verification, notification services, rental conflict validation, inventory management, and business analytics into a single ecosystem.

5 System Design

5.1 Use Case Design

Main actors:

- Guest
- Customer
- Owner
- Staff
- Administrator

The use case diagrams for each actor are presented in Figures 1–6.

Figure 7 presents the Entity Relationship Diagram (ERD) of the proposed system.

6 Implementation

6.1 Technologies

- ReactJS
- React Native
- ASP.NET Core
- MongoDB
- GitHub
- Postman

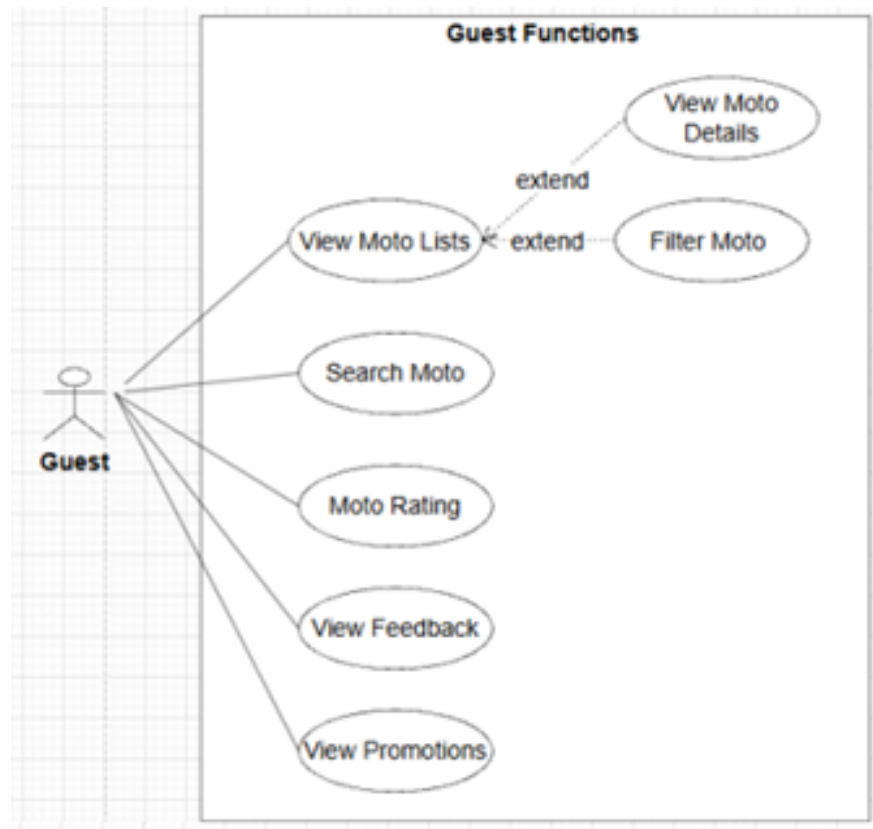


Fig. 1. Guest Use Case Diagram

6.2 Current Progress

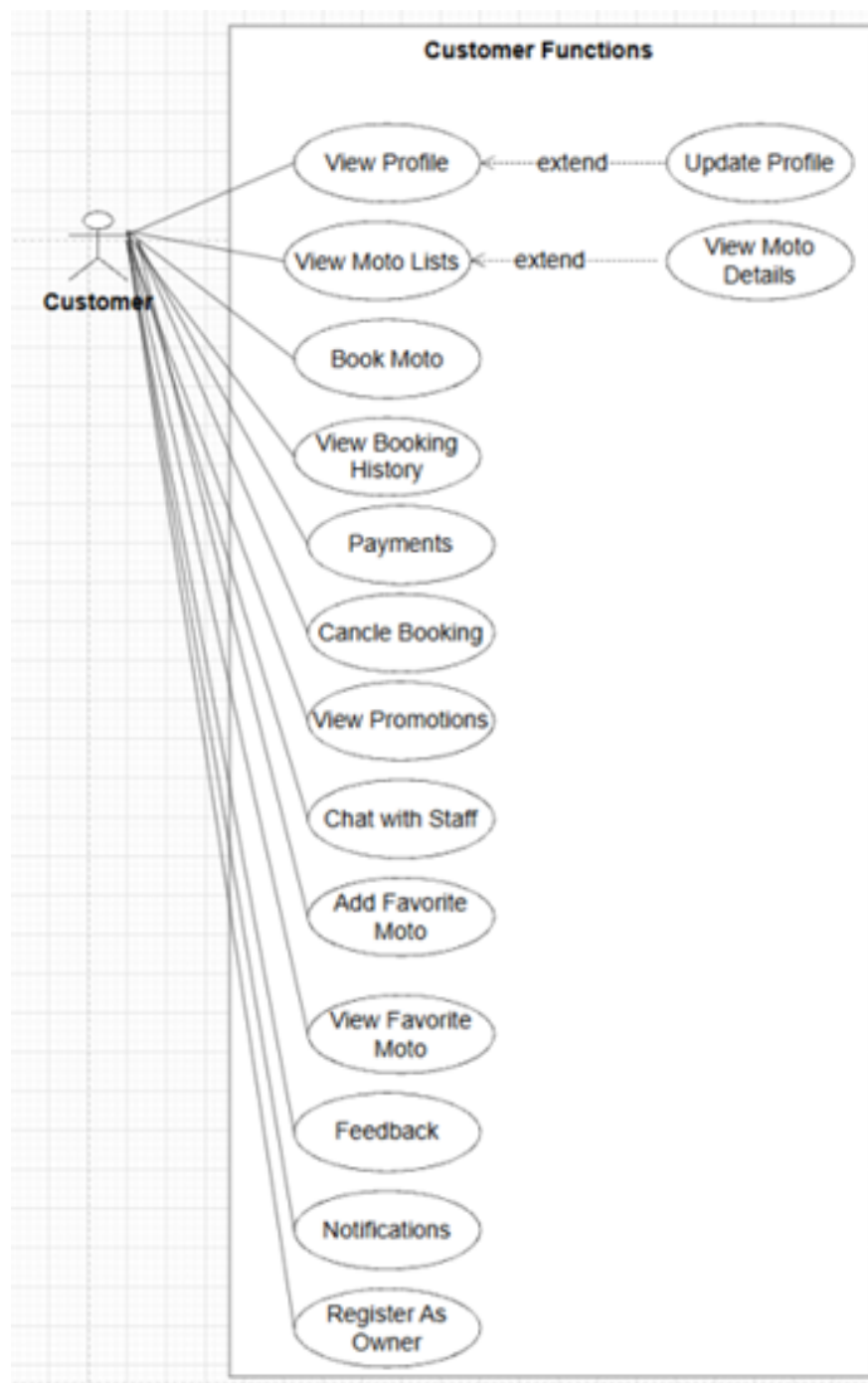
- Requirements Analysis
- Literature Review
- Use Case Design
- Database Design (ERD)
- User Interface Design
- Authentication Module
- Google Authentication Integration
- Citizen ID Verification Design
- Booking Management Module
- Rental Conflict Validation Logic
- Rental Price Calculation Module
- Payment Integration Design
- Notification Service Design
- CRUD Functions for Users, Vehicles, and Bookings

7 Conclusion

This paper presented Motov, a hybrid motorbike rental platform integrating company-owned and partner-owned vehicles into a unified rental ecosystem. The proposed system supports customers, partners, and administrators through web and mobile applications. Future work includes recommendation systems, real-time vehicle tracking, and advanced analytics.

References

1. Andersen, A.R.: A queueing model for time-dependent rental systems with phase-type distributed rentals and substitutions of items. *Annals of Operations Research* **350**(3), 1143–1167 (Jul 2025). <https://doi.org/10.1007/s10479-025-06645-1>, <https://link.springer.com/10.1007/s10479-025-06645-1>
2. González, A.B.R., Wilby, M.R., Díaz, J.J.V., Pozo, R.F., Ávila, C.S.: Utilization rate of the fleet: a novel performance metric for a novel shared mobility. *Transportation* **50**(1), 285–301 (Feb 2023). <https://doi.org/10.1007/s11116-021-10244-x>, <https://link.springer.com/10.1007/s11116-021-10244-x>
3. Khwaja, A., Yang, N.: Quantifying the link between employee engagement, and customer satisfaction and retention in the car rental industry. *Quantitative Marketing and Economics* **20**(3), 275–292 (Sep 2022). <https://doi.org/10.1007/s11129-022-09253-6>, <https://link.springer.com/10.1007/s11129-022-09253-6>
4. Liu, H., Luo, K., Xu, Y., Zhang, H.: Online car-sharing problem with variable booking times. *Journal of Combinatorial Optimization* **47**(3), 32 (Apr 2024). <https://doi.org/10.1007/s10878-024-01114-0>, <https://link.springer.com/10.1007/s10878-024-01114-0>
5. Münzel, K., Boon, W., Frenken, K., Vaskelainen, T.: Carsharing business models in Germany: characteristics, success and future prospects. *Information Systems and e-Business Management* **16**(2), 271–291 (May 2018). <https://doi.org/10.1007/s10257-017-0355-x>, <http://link.springer.com/10.1007/s10257-017-0355-x>
6. Neifer, T., Bossauer, P., Pakusch, C., Boehm, L., Lawo, D.: Trust-Building in Peer-to-Peer Carsharing: Design Case Study for Algorithm-Based Reputation Systems. *Computer Supported Cooperative Work (CSCW)* **33**(2), 137–171 (Jun 2024). <https://doi.org/10.1007/s10606-022-09461-4>, <https://link.springer.com/10.1007/s10606-022-09461-4>
7. Soppert, M., Oliveira, B.B., Angeles, R., Steinhardt, C.: On the benefit of combining car rental and car sharing. *Journal of Business Economics* **94**(9), 1261–1298 (Nov 2024). <https://doi.org/10.1007/s11573-024-01204-9>, <https://doi.org/10.1007/s11573-024-01204-9>

**Fig. 2.** Customer Use Case Diagram

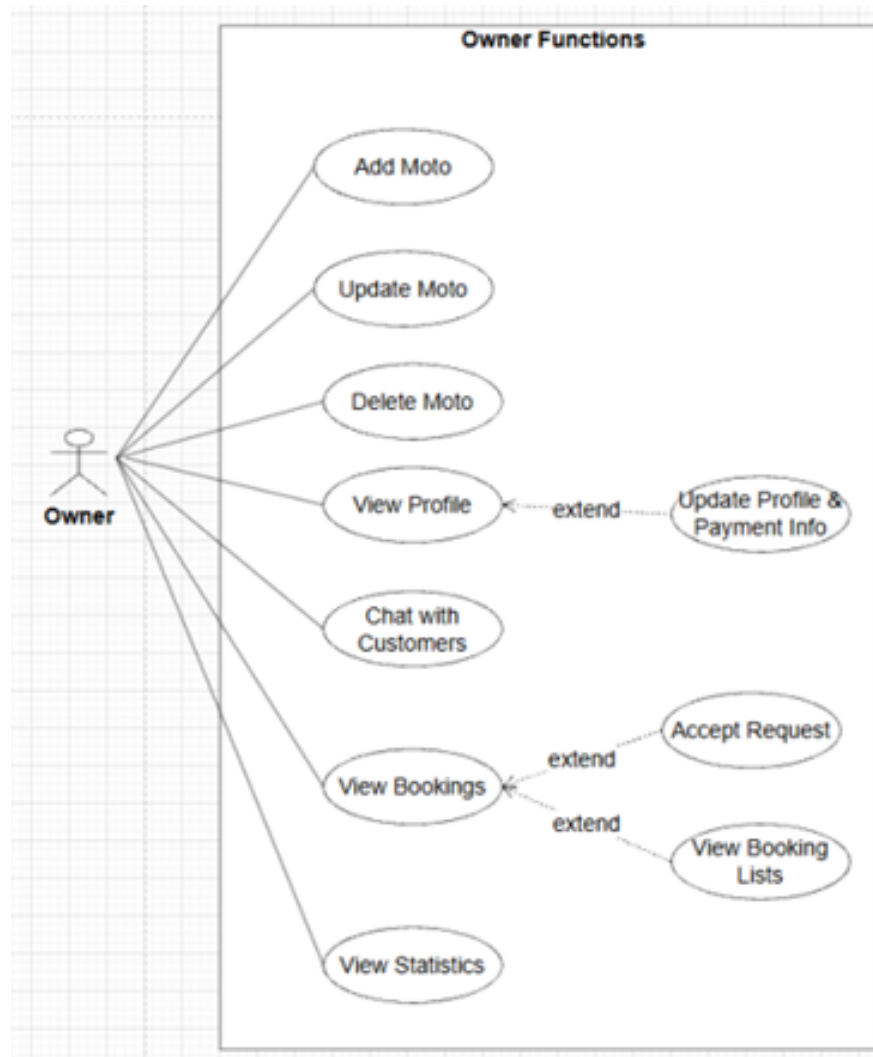


Fig. 3. Owner Use Case Diagram

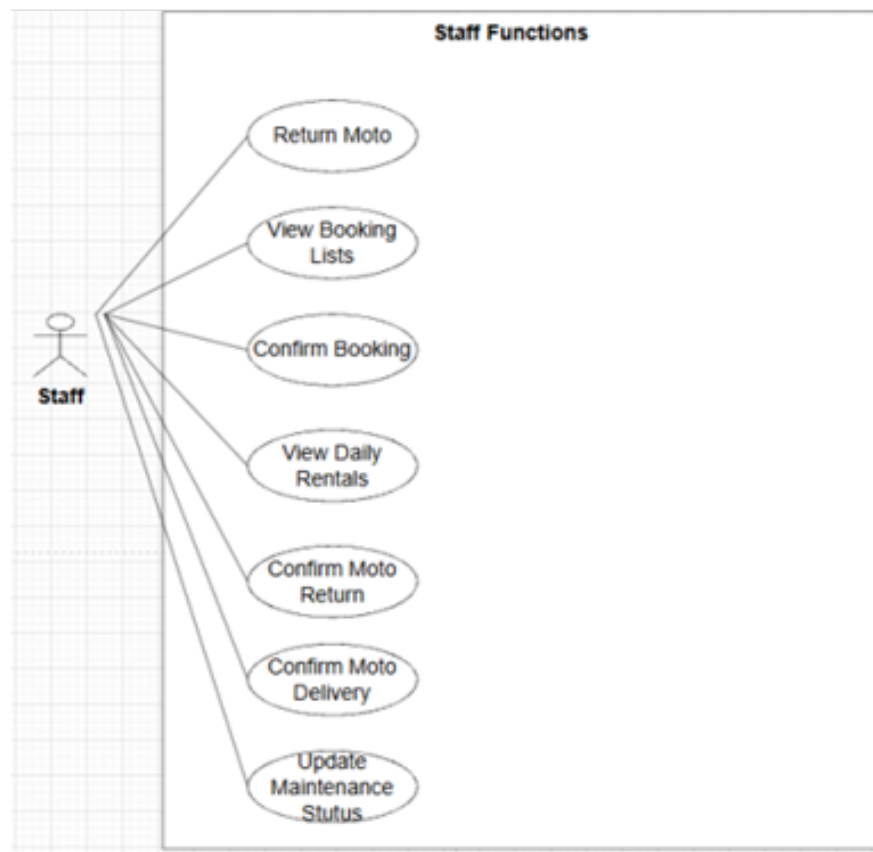


Fig. 4. Staff Use Case Diagram

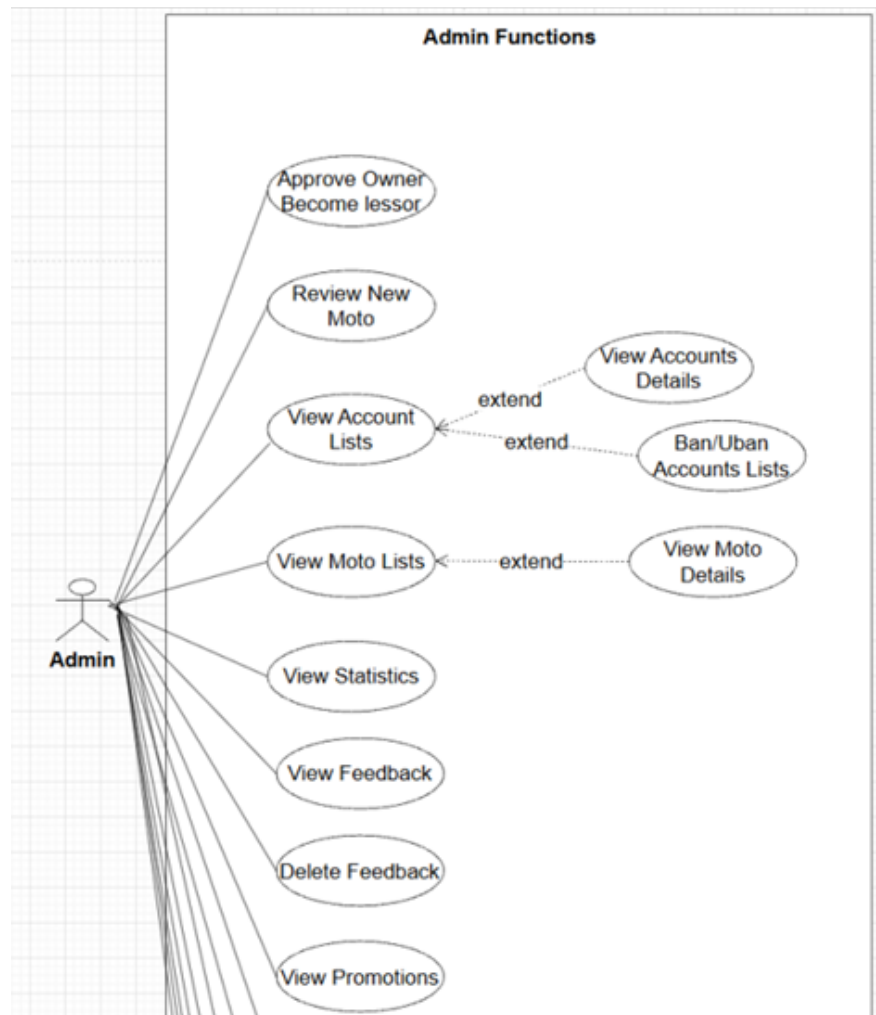
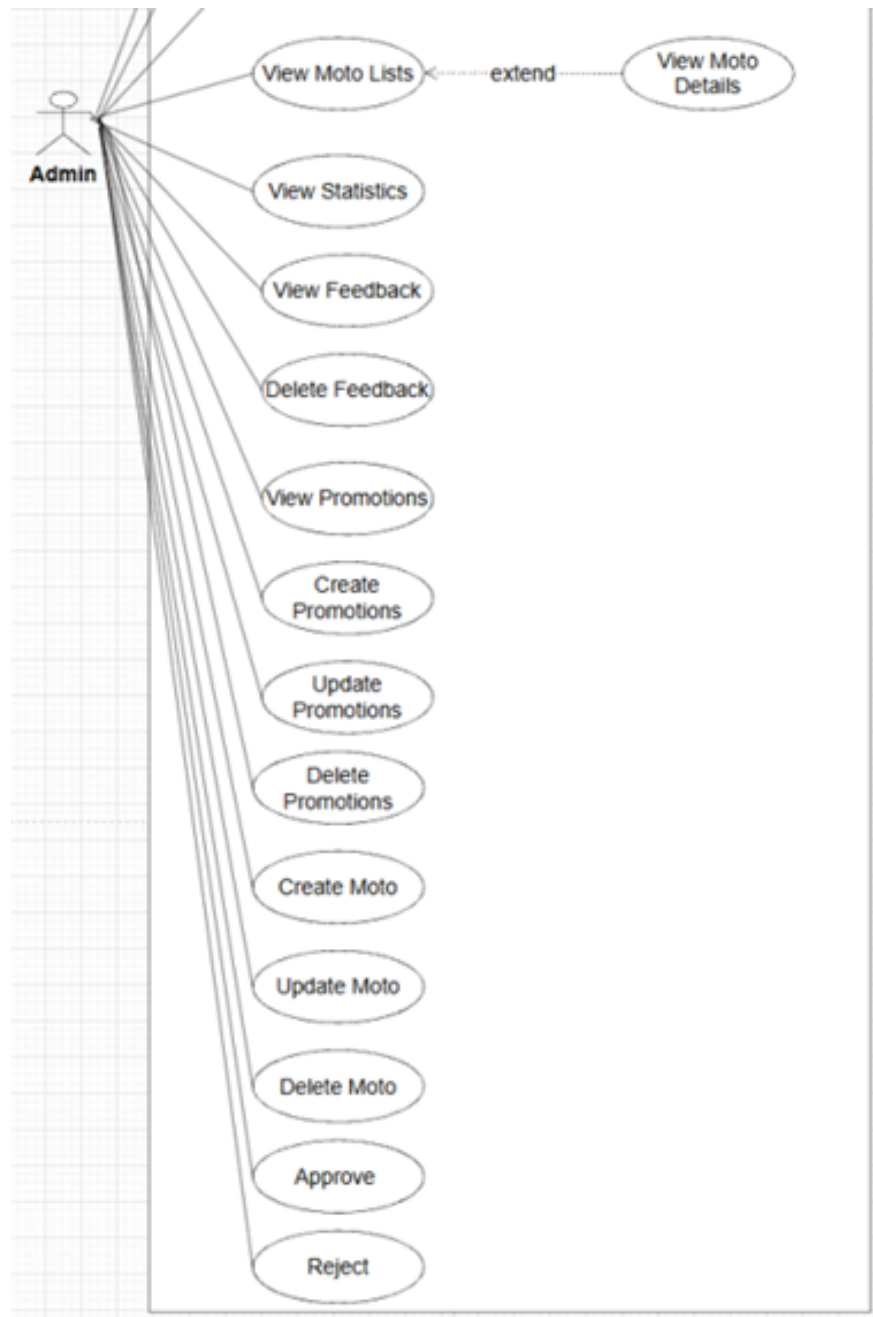


Fig. 5. Administrator Use Case Diagram (Part 1)

**Fig. 6.** Administrator Use Case Diagram (Part 2)

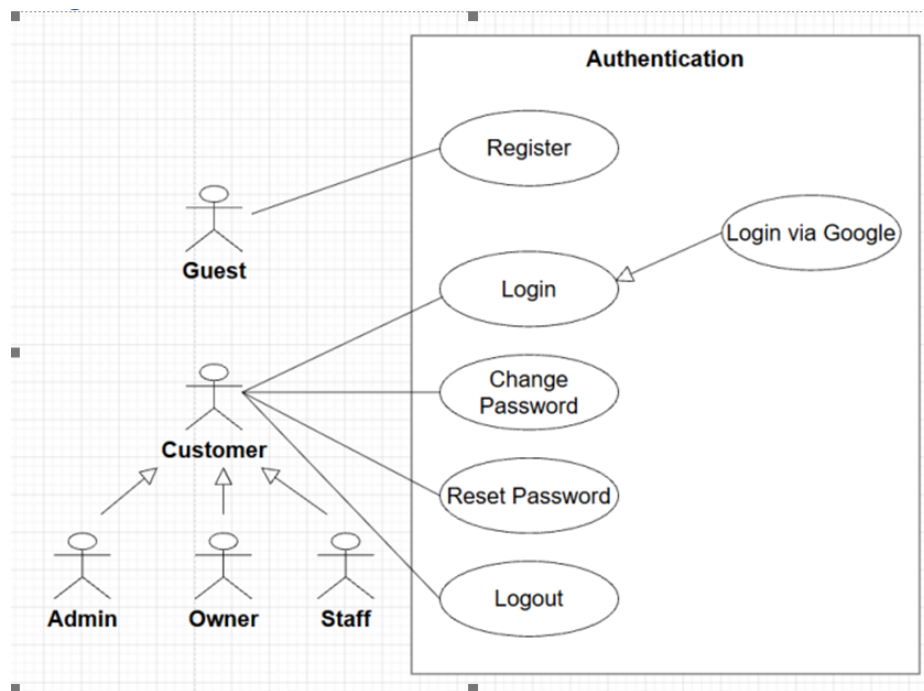


Fig. 7. Entity Relationship Diagram of the Motov system